

Test 3

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Question 1:

1. Since 101 is prime, $\mathbb{Z}/101\mathbb{Z}$ forms a field because every non-zero element has a multiplicative inverse. Yes, $(\mathbb{Z}/101\mathbb{Z}, +, \cdot)$ is a field.
2. Since 100 is not prime, $\mathbb{Z}/100\mathbb{Z}$ is not a field. For instance, the element 2 does not have a multiplicative inverse modulo 100 because $\gcd(2, 100) \neq 1$. No, $(\mathbb{Z}/100\mathbb{Z}, +, \cdot)$ is not a field. Example of a non-invertible element: 2.
3. This is the set of all 3×3 matrices with rational entries. Not all matrices have inverses (eg. singular matrices), so this set does not form a field. No, $\mathbb{Q}_{3 \times 3}$ is not a field. Example of a non-invertible element: any singular 3×3 matrix (like one with all zeros).

Question 2:

1. $\text{Span}_{\mathbb{R}} \{(1, 1, 1, 1), (1, 0, 0, 1), (0, 0, 0, 2), (2, 12, 0, 0)\}$
The dimension is 4, since the vectors are linearly independent.
2. $\mathbb{Q}^3$: This is the space of 3-dimensional vectors with rational entries. Dimension is 3.
3. $\mathbb{Q}_{3 \times 5}$: This is the space of 3×5 matrices with rational entries. Dimension is $3 \times 5 = 15$.

Question 3:

$$\{1, 4, 3, 5, 2\} \rightarrow \{1, 2, 3, 5, 4\} \rightarrow \{1, 2, 3, 4, 5\}$$

We obtain reordering by swapping 2 and 4, followed by 4 and 5. $\therefore$ the signature is 1.

Question 4:

$$1. A = \begin{bmatrix} 9 & 1 & -4 \\ -5 & 6 & -10 \\ -10 & -4 & -4 \end{bmatrix}$$

Expanding with respect to 1st row:

$$(-1)^{1+1} \cdot (9) \cdot \begin{vmatrix} 6 & -10 \\ -4 & -4 \end{vmatrix} + (-1)^{1+2} \cdot (1) \cdot \begin{vmatrix} -5 & -10 \\ -10 & -4 \end{vmatrix} + (-1)^{1+3} \cdot (-4) \cdot \begin{vmatrix} -5 & 6 \\ -10 & -4 \end{vmatrix}$$

$$= 9(-24 - 40) - (20 - 100) - 4(-20 + 60)$$

$$= -576 - 80 - 320$$

$$= -816$$

$$2. C_{11} = (-1)^{1+1} \begin{vmatrix} 6 & -10 \\ -4 & -4 \end{vmatrix} = -64$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} -5 & -10 \\ -10 & -4 \end{vmatrix} = 80$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} -5 & 6 \\ -10 & -4 \end{vmatrix} = 80$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 1 & -4 \\ -4 & -4 \end{vmatrix} = 20$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 9 & -4 \\ -10 & -4 \end{vmatrix} = -76$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 9 & 1 \\ -10 & -4 \end{vmatrix} = 26$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 1 & -4 \\ 6 & -10 \end{vmatrix} = 14$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 9 & -4 \\ -5 & -10 \end{vmatrix} = 110$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 9 & 1 \\ -5 & 6 \end{vmatrix} = 59$$

$$C = \begin{bmatrix} -64 & 80 & 80 \\ 20 & -76 & 26 \\ 14 & 110 & 59 \end{bmatrix}$$

3. Inverse of A is

$$A^{-1} = \frac{1}{\det(A)} C^T = \frac{1}{-816} \begin{bmatrix} -64 & 20 & 14 \\ 80 & -76 & 110 \\ 80 & 26 & 59 \end{bmatrix}$$

$$= \begin{bmatrix} 4/51 & -5/204 & -7/408 \\ -5/51 & 19/204 & -55/408 \\ -5/51 & -13/408 & -59/816 \end{bmatrix}$$

Question 5: $A = \begin{bmatrix} 2 & -14 \\ -16 & 114 \end{bmatrix}$ $A^T = \begin{bmatrix} 2 & -16 \\ -14 & 114 \end{bmatrix}$

$$A^T \cdot A = \begin{bmatrix} (2) \cdot (2) + (-16) \cdot (-16) & (2) \cdot (-14) + (-16) \cdot (114) \\ (-14) \cdot (2) + (114) \cdot (-16) & (-14) \cdot (-14) + (114) \cdot (114) \end{bmatrix}$$

$$= \begin{bmatrix} 260 & -1852 \\ -1852 & 13192 \end{bmatrix}$$

$$\det(A^T A) = (260) \cdot (13192) - (-1852) \cdot (-1852) = 16$$

$$\text{Vol}(\mathcal{L}) = \sqrt{\det(A^T A)} = \sqrt{16} = 4$$

$$\begin{bmatrix} 2 & -14 \\ -16 & 114 \end{bmatrix} \xrightarrow{C_2 \rightarrow C_2 + 7C_1} \begin{bmatrix} 2 & 0 \\ -16 & 2 \end{bmatrix} \xrightarrow{C_1 \rightarrow C_1 + 8C_2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\lambda_1(\mathcal{L}) = \sqrt{2}$$

$$\lambda_2(\mathcal{L}) = \sqrt{2}$$

Question 6: The matrix would be $A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -2 \\ 2 & 0 & 1 \end{bmatrix}$

Question 7: $A_x = (5 + 2x \quad 9x^2 + x + 5)$
 $B_x = (10x^2 + x + 9)$
 $S = \begin{pmatrix} x+1 \\ x-1 \end{pmatrix}$

$\text{buff} = (5x^2 + x + 7) \leftarrow \text{Calculation done on Maple}$

Since, 5 and 7 are close to $q/2$, therefore they will be 1 and m , will be 0 since it is closer to 0.
 Therefore the message is $x^2 + 1$.

Question 8: Calculations done on Maple

$e = \begin{bmatrix} 0 \\ -1 \\ -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad S = \begin{bmatrix} 27 \\ 87 \\ 99 \end{bmatrix}$